Project Interview Preparation Guide

Project Interview Preparation Guide

General Project-Based Interview Questions and Answers

1. Can you briefly describe the projects you have worked on?

Answer:

Yes, Ive worked on two major projects:

1. Real-Time Road Lane Detection using OpenCV and Python This project uses computer vision techniques to identify and highlight lane markings from video feed.

2. Walmart Sales Data Analysis using SQL and Python This is an end-to-end data analytics project using PostgreSQL for SQL querying and Python for data processing.

2. What technologies or tools did you use in each project?

Answer:

- For the Road Lane Detection project: Python, OpenCV.
- For the Walmart Analysis project: SQL (PostgreSQL), Python (Pandas, Matplotlib), GitHub.

3. What challenges did you face and how did you solve them?

Answer:

- In Lane Detection: Handling poor lighting, curves, and shadows. I used curve fitting and image preprocessing techniques like Gaussian blur and Canny edge detection.
- In Walmart Analysis: Large dataset cleaning and query optimization. I handled NULLs and duplicates using SQL and created optimized joins and filters.

4. How did you ensure the reliability or accuracy of your system?

Answer:

- In lane detection, I tested on various road types and lighting conditions.
- In sales data analysis, I validated SQL query outputs with visual insights in Python.

5. How did you manage version control?

Project Interview Preparation Guide

Answer:

I used GitHub to manage files and track changes for the Walmart project. I committed updates, deletions, and documentations regularly.

6. Did you document your work?

Answer:

Yes, for the Walmart project, I documented all findings and SQL queries on GitHub to support collaboration and project transparency.

7. How does your project contribute to real-world applications?

Answer:

- The lane detection project can be used in ADAS (Advanced Driver Assistance Systems) to help cars identify road boundaries.
- The Walmart project delivers business insights on sales trends and profitability for retail optimization.

Detailed Project Descriptions

1. Real-Time Road Lane Detection, Python, OpenCV

- Developed a real-time lane detection system using OpenCV in Python.
- Used Gaussian blur, Canny edge detection, region masking, and Hough transform to detect lane lines.
- Implemented curve fitting to accurately detect lanes on curved roads.
- Handled challenges such as low light and shadow by adaptive thresholding and preprocessing.
- Goal: Assist driver systems in identifying road lanes under various conditions.

2. Walmart Data Analysis Project, SQL + Python, PostgreSQL

- Developed a complete data analysis pipeline for Walmart sales data.
- Extracted, cleaned, and transformed data using PostgreSQL queries.
- Identified sales trends, top-performing products, and profitability metrics.
- Cleaned data by removing duplicates and NULLs, and restructured schema.
- Used Python to visualize and explore insights.
- Managed files and documentation on GitHub.